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Postdoctoral Fellow

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Education

- 2004/09 – 2010/06: **Ph.D.** Institute of Geophysics, National Central University (NCU), Taiwan
- 1999/09 – 2001/06: **M.S.** Institute of Geophysics, NCU, Taiwan
- 1995/09 – 1999/06: **B.S.** Department of Oceanography, National Taiwan Ocean University, Taiwan

Experience

- 2022/03 – present: Postdoctoral fellow, Center for Environmental Studies (CES), NCU, Taiwan
- 2018/09 – 2021/12: Associate professor, College of Oceanography, Hohai University (HHU), Nanjing, China
- 2019/02 – 2020/02: Associate Head, Department of Marine Geology and Geophysics, HHU, Nanjing, China
- 2016/08 – 2018/08: Postdoctoral fellow, CES/Department of Earth Sciences, NCU, Taiwan
- 2010/09 – 2016/07: Postdoctoral fellow, Institute of Earth Sciences, Academia Sinica, Taiwan
- 2008/05 – 2009/10: Visiting student researcher**,** Pacific Geosciences Centre, Geological Survey of Canada, Canada
- 2004/09 – 2010/06: Graduate research assistant, Institute of Geophysics, NCU, Taiwan
- 2003/07 – 2004/08: Research assistant, Institute of Geophysics, NCU, Taiwan

- 2001/10 – 2003/06: Artillery Second Lieutenant, compulsory military service, ROC Army
- 1999/09 – 2001/06: Teaching assistant, Institute of Geophysics, NCU, Taiwan

Current Research Interests

- Delineating seismogenic structures offshore and onshore Taiwan for providing critical constraints on Taiwan orogeny and seismic hazard assessment.
- Revealing earthquake source characteristics of subduction zone earthquakes and intraplate events for better understanding of mechanisms of earthquake nucleation and triggering, now focus on the western boundary of the Philippine Sea Plate.
- Analyzing seismicity data for comprehending earthquake processes and lithosphere deformation.

Teaching at Hohai University (2018/09 – 2021/12；outstanding teaching award, 2018-2019; 2019-2020)

- The origin of continents and oceans: Introduction to Plate Tectonics (Literacy class; Fall, 2018 and 2019)
- Marine environment, resource and climate change: Introduction to Geophysics (Introductory course for College of Oceanography freshman; Fall, 2018 and 2019)
- Introduction to geology (Undergraduate course, Fall, 2018 and 2019)
- Volcanology: Introduction to Volcano Seismology (Undergraduate course, Spring, 2019)
- Geophysics: Introduction to Seismology (Undergraduate course, Spring, 2019, 2020 and 2021)
- Marine Tectonics and Geophysics: Introduction to Seismology and Earth Structure (Graduate course; Spring, 2019, 2020 and 2021)
- Undergraduate Research (Fall, 2019)
- Undergraduate student mentor (Fall, 2019 – Fall, 2021)

Peer-Reviewed Publication

23. He, X-B., **W.-N. Wu**, T.-T. Dou and J.-T. Tao (2023), Rapid report of the 27 July 2022 Ms 7.0 Luzon earthquake, Philippines, *Earthquake Research Advances*, in revision.

22. Doo, W.-B., C.-L. Lo, Y.-S. Huang, **W.-N.** Wu, and S.-Y. Wang (2023), Variations in the crustal structure and strength of plate coupling along the Ryukyu subduction zone, *Geoscience Letters*, in revision.

21. Wu, W.-N, C.-L. Lo, W-B Doo, J-Y Lin, and S.-K. Hsu (2021), Seismogenic structure along the deformation front off SW Taiwan revealed by the aftershocks of the 2017 Tainan near-shore earthquake with ocean bottom seismometers, *Tectonophysics*, 815, 228995, doi:10.1016/j.tecto.2021.228995.

20. Doo, W.-B., **W.-N. Wu**, Y.-S. Huang, C.-L. Lo, H.-F. Wang, S.-Y. Wang, and H. Kuo-Chen (2021), Deep crustal structure in the Taiwan–Ryukyu arc–trench system junction area: New constraints from gravity modelling, *Terra Nova*, 33, 407-414, doi:10.1111/ter.12525.

19. Doo, W.-B., C.-L. Lo, H. Kuo-Chen, Y.-S. Huang, **W.-N. Wu**, S.-K. Hsu and H.-F. Wang (2020), Variations in mantle lithosphere buoyancy reveal seismogenic behavior in the Sunda-Andaman subduction zone, *Geophysical Journal International*, 220, 2, 1275-1283, doi:10.1093/gji/ggz502.

18. Lo, C.-L., **W.-N. Wu**, W.-B. Doo, Y.-S. Huang, and S.-K. Hsu (2019), Crustal stress field in Ilan Plain, NE Taiwan and the westernmost Okinawa trough- deduced from seismic stress inversion, *Terrestrial Atmosphere Oceanic Sciences*, 30, 5, 613-619, doi:10.3319/TAO.2019.08.09.01.

17. Lin, J.-Y., **W.-N. Wu**, C.-C. Su, C.-H. Tsai, H.-H. Sun, Y.-F. Chen, S.-J. Chin, and Y.-C. Lin (2019), Spatial and temporal distribution of low frequency volcanic earthquakes in the southern Okinawa Trough back-arc basin, *Terrestrial Atmosphere Oceanic Sciences*, 30, 5, 621-631, doi:10.3319/TAO.2018.12.24.01.

16. Doo, W.-B., C.-L. Lo, **W.-N. Wu**, J.-Y. Lin, S.-K., Hsu, Y.-S., Huang, and H.-F. Wang, (2018), Strength of plate coupling in the southern Ryukyu subduction zone, *Tectonophysics*, 723, 223-228, doi:10.1016/j.tecto.2017.12.028.

15. Lin, J.-Y., Y.-F. Chen, C.-C. Su, S.-J., Chin, W.-B., Cheng, **W.-N. Wu**, C.-W. Liang, H.-S. Hsieh, S.-K. Hsu, and Y.-C. Lin (2018), Seismic site response of submarine slope offshore southwestern Taiwan, *Terrestrial Atmosphere Oceanic Sciences*, 29, 1, 51-63, doi:10.3319/TAO.2017.05.09.01.

14. Wu, W.-N., Y.-T. Yen, Y.-J. Hsu, Y.-M. Wu, J.-Y. Lin and S.-K. Hsu (2017), Spatial variation of the seismogenic depth

of crustal earthquakes in the Taiwan region and its implications on seismic hazard assessment, *Tectonophysics*, 708, 81-95, doi:10.1016/j.tecto.2017.04.028.

13. Wu, W.-N., C.-L. Lo and J.-Y. Lin (2017), Spatial variation of the crustal stress field in the Philippine region and its tectonic implications, *Journal of Asian Earth Sciences*, 142, 109-118, doi:10.1016/j.jseaes.2017.01.036.

12. Chin, S.-J., J.-Y. Lin, Y.-F. Chen, **W.-N. Wu** and C.-W. Liang (2016), Transition of the Taiwan-Ryukyu collision-subduction process as revealed by ocean-bottom seismometer, *Journal of Asian Earth Sciences*, 128, 149-157, doi:10.1016/j.jseaes.2016.07.008.

11. Wu, W.-N. (2015), Comment on the paper by Kavitha and Raghukanth "Regional level forecasting of seismic energy release", *Acta Geod Geophys*, 51, 773-775, doi:10.1007/s40328-015-0157-x.

10. Lin, J.-Y., **W.-N. Wu** and C.-L. Lo (2015), Megathrust earthquake potential of the Manila subduction systems revealed by the radial component of seismic moment tensor Mrr, *Terrestrial Atmosphere Oceanic Sciences*, 26, 619-630, doi:10.3319/TAO.2013.04.29.01(TC).

9. Lin, J.-Y., C.-L. Lo, **W.-N. Wu**, J.-C. Sibuet, S.-K. Hsu and Y.-Y. Wen (2015), Crustal thickening and extension induced by the Great Sumatra-Andaman earthquake of 26 December 2004: revealed by the seismic moment tensor element Mrr, *Marine Geophysical Research*, 36, 187-195, doi:10.1186/1880-5981-66-49.

8. Lin, J.-Y., J.-C. Sibuet, S.-K. Hsu, and **W.-N. Wu** (2014), Could a Sumatra-like megathrust earthquake occur in the south Ryukyu subduction zone? *Earth Planets and Space*, 66:49, doi:10.1186/1880-5981-66-49.

7. Wu, W.-N. and L. Zhao (2013), Delineating seismogenic structures by a revised best estimate method: Application to the Taiwan orogenic belt, *Journal of Seismology*, 17, 545-556, doi. 10.1007/s10950-012-9336-1.

6. Wu, W.-N., L. Zhao, and Y.-M. Wu (2013), Empirical relationships between aftershock zone dimensions and moment magnitudes for plate boundary earthquakes in Taiwan, *Bulletin of the Seismological Society of America*, 103, doi: 10.1785/0120120173.

5. D'Amico S., B. Orecchio, D. Presti, G. Neri, **W.-N. Wu**, I. Sandu, L. Zhu, and R. B. Hermann (2013), Source parameters

of small and moderate earthquakes in the area of the 2009 L'Aquila earthquake sequences (central Italy), *Physics and Chemistry of the Earth*, 63, 77-91, doi:10.1016/j.pce.2013.02.005.

4. Lin, J.-Y. and **W.-N. Wu** (2012), Spatio-temporal distribution of seismic moment release near the source area of the 2011 Tohoku-Oki earthquake, *Earth Planets and Space*, 64, 1067-1075, doi:10.5047/eps.2012.04.006.

3. Armada, L., S.-K. Hsu, C.-Y. Ku, W.-B. Doo, **W.-N. Wu**, C. Dimalanta, and G. Yumul (2012), Possible northward extension of the Philippine Fault Zone offshore Luzon Island (Philippines), *Marine Geophysical Research*, 33, 369-377, doi:10.1007/s11001-013-9169-5.

2. Wu, W.-N., H. Kao, S.-K. Hsu, C.-L. Lo, and H.-W. Chen (2010), Spatial variation of the crustal stress field along the Ryukyu-Taiwan-Luzon convergent boundary, *Journal of Geophysical Research: Solid Earth*, 115, B11401, doi:10.1029/2009JB007080.

1. Wu, W.-N., S.-K. Hsu, C.-L. Lo, H.-W. Chen, and K.-F. Ma (2009), Plate convergence at the westernmost Philippine Sea Plate, *Tectonophysics*, 466, 162-169, doi:10.1016/j.tecto.2007.11.011.

Field Experience

- Research cruise, Ocean Bottom Seismometer (OBS) recovery South China Sea (03/2023; 22 days)
- Research cruise, OBS recovery off SW Taiwan (03/2017; 2 days)

Academic Service and Synergistic Activity

- Served as reviewer for *BSSA, EPS, GRL, GRAPY, GSAB, JAES, JGR-Solid, SRL, PAG, TAO, Tectonophysics*.
- Judge of Outstanding Student Paper Awards: Taiwan Geosciences Assembly **(**2011-2017)
- Serve in Editor Board: Member of the Editorial Board *Mistral Service Press* (2015 - present)
- Served as Consultant

- National Standards Review Committee number of Ocean Bottom Seismograph, Ministry of Natural Resources, China (2019)

- Workshop on Taiwan SSHAC Level 3 Project, invited by the National Center for Research on Earthquake Engineering, Taipei, 2017.
- Workshop on Seismogenesis of the Subduction Zone in South Taiwan, invited by the Institute of Nuclear Energy Research and the Institute of Geophysics, National Central University, Jhongli, 2014.
- Workshop on Seismogenesis of the Subduction Zone in North Taiwan, invited by the Institute of Nuclear Energy Research and SINOTECH engineering consultants inc., Taipei, 2013.

Honor and Fellowship

- Academia Sinica Postdoctoral Research Fellowship, 2011-2012
- Outstanding Graduate Research Scholarship, National Central University, 2008
- Recipient of the Graduate Students Study Abroad Program sponsored by National Science Council, 2008

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